

L4.1 Adversarial Attacks Quiz

1. A traffic-sign classifier is trained on sunny daytime photos and deployed in heavy snow, where accuracy drops. How does the lecture classify this?

- A) A white-box adversarial attack, because the snow changes pixel values inside an ℓ_∞ norm ball
- B) A targeted attack, because the sign is always misread as one specific wrong class by the model
- C) A naturally occurring distribution shift: no attacker is needed, yet the i.i.d. assumption fails
- D) A black-box attack, because the weather conditions are unknown to the designer of the model

ANS:

2. For the perturbation $\delta = (3, -4, 12)$, which triple gives $(\ell_1, \ell_2, \ell_\infty)$?

- A) (11, 13, 12)
- B) (19, 13, 12)
- C) (19, 169, 12)
- D) (19, 13, 19)

ANS:

3. Perturbation A changes two pixels by 7 each, $\delta = (7, 7)$; perturbation B changes one pixel by 10, $\delta = (10, 0)$. Which comparison is correct?

- A) The ℓ_∞ sizes are equal at 10, while ℓ_2 ranks perturbation A as clearly the larger of the two
- B) ℓ_2 sizes are nearly equal (≈ 9.9 vs 10), but ℓ_∞ is 7 vs 10, so B looks far larger under ℓ_∞
- C) Perturbation A is larger under both norms, because it changes more pixels than B does here
- D) The ℓ_2 sizes are 14 vs 10, so A is larger, and ℓ_∞ cannot be computed for a single pixel

ANS:

4. What is the ℓ_∞ ball $\{\delta : \|\delta\|_\infty \leq \epsilon\}$ for a 2-pixel image, and why does this matter for FGSM?

- A) A circle of radius ϵ ; FGSM moves along the radius that points toward the gradient direction
- B) A diamond with corners at $\pm\epsilon$ on each axis; FGSM changes only the pixel with the largest gradient
- C) A line segment of length ϵ ; FGSM can change only one pixel by one step of that size
- D) A square with side 2ϵ ; its corners move every pixel by exactly $\pm\epsilon$, which is where FGSM ends up

ANS:

5. Model training and adversarial example generation both use gradients. How do their updates differ?

- A) Training ascends the loss in θ ; the attack descends the loss in δ so the output matches the label
- B) Training descends the loss in θ with x fixed; the attack ascends the loss in δ with θ fixed
- C) Both descend the loss, but the attack uses a much larger learning rate and updates θ as well
- D) Training updates the input x to fit the labels; the attack updates θ so that the loss grows

ANS:

6. In PGD, with respect to which variable is the gradient of the loss taken, and why?

- A) With respect to δ : the image x is a constant, and the perturbation is the optimization variable
- B) With respect to θ , because the attacker retrains the network a little at every step of the loop
- C) With respect to the label y , because the attacker moves the label toward the chosen target class
- D) With respect to nothing, because only the sign of the loss value is used in each iteration here

ANS:

7. FGSM with $\epsilon = 0.1$. The loss gradient at $\delta = 0$ is $(-3.0, 0.5, 0.0)$. What is the FGSM perturbation δ ?

- A) $(-0.3, +0.05, 0)$
- B) $(-0.1, +0.1, +0.1)$
- C) $(-0.1, +0.1, 0)$
- D) $(+0.1, -0.1, 0)$

ANS:

8. Why does the exact magnitude of the gradient not matter for FGSM?

- A) A very large step is clipped back to $\pm\epsilon$ in each coordinate, so only the gradient sign survives
- B) The loss is linear in δ , so the gradient magnitude cancels out inside the softmax operator at once

- C) The gradient is always normalized to unit ℓ_2 length before any update is applied to the vector δ
- D) The magnitude matters only for targeted attacks and is irrelevant for an untargeted attack here

ANS:

9. PGD uses $\epsilon = 8/255$ and a step size $\alpha = 2/255$. How many iterations does the rule of thumb suggest, and why not just one?

- A) A few times $\epsilon/\alpha = 4$, e.g. 10 to 20; one step of size α cannot reach the ball's boundary
- B) Exactly one, because the single-step FGSM direction is already the strongest attack possible
- C) About 255, one iteration for each of the possible pixel intensity levels of the image data
- D) Exactly two, because α is $2/255$ and each iteration must change two pixels at a time by α

ANS:

10. Compared with FGSM at the same ϵ , what does PGD with small steps typically achieve?

- A) The same loss with a smaller perturbation, using one gradient evaluation like FGSM does here
- B) A perturbation that leaves the ϵ -ball, since projection is applied only once at the very end
- C) A higher loss inside the same ϵ -ball, paid for with several gradient evaluations, not one
- D) Exactly the same δ as FGSM, because the projection at each step erases all of the differences

ANS:

11. The true class is the first one. The logits change from $(2, 0, 0)$ to $(0, 2, 0)$. How does the cross-entropy loss, $\log \sum \exp(h_j) - h_y$, change?

- A) It falls from about 2.24 to about 0.24, because the largest logit now sits at the wrong class
- B) It rises from 0 to 2, because the softmax outputs sum to 2 after the logits are exchanged
- C) It rises from about 0.24 to about 2.24: the log-sum-exp is unchanged, but $-h_y$ grows by 2
- D) It stays at about 0.24, because the sum of the logits did not change after the exchange

ANS:

12. Which objective, written as a minimization over δ , describes a targeted attack toward class t ?

- A) Minimize $h_y(x+\delta)$ alone: only the true-class logit matters, and the target comes out by luck
- B) Maximize $h_y(x+\delta) - h_t(x+\delta)$: raise the true-class logit relative to the target-class logit
- C) Minimize $\|\delta\|$ with no reference to any logit: the smallest perturbation is always sufficient
- D) Minimize $h_y(x+\delta) - h_t(x+\delta)$: lower the true-class logit while raising the target-class logit

ANS:

13. In the physical stop-sign attack, why is the loss averaged over k transformed copies of the sign image?

- A) To estimate the gradient of a black-box model from the noisy queries sent to an online service
- B) So one δ keeps working across the viewing angles, distances and lighting that a camera will see
- C) To shrink the ℓ_p norm of δ , because averaging cancels out most of the individual pixel changes
- D) To enforce the ℓ_∞ bound without a projection step, because averaging clips each pixel value

ANS:

14. Why is an ℓ_1 term on $M_x \cdot \delta$ used when the perturbation is localized to parts of the sign?

- A) ℓ_1 equals ℓ_∞ for a binary mask, so the ϵ bound is enforced exactly in every masked pixel here
- B) ℓ_1 promotes sparsity, so the change concentrates in a few of the most vulnerable regions
- C) ℓ_1 penalizes large single pixels most, so the change is spread evenly over the whole sign area
- D) ℓ_1 makes the objective differentiable, whereas ℓ_2 would make the loss non-differentiable there

ANS:

15. What does a robust model's loss surface look like, according to the lecture, and why does it help the defender?

- A) Flatter because capacity is reduced; fewer parameters mean fewer directions for an attacker to use
- B) Non-differentiable, so no gradient exists and gradient-based attackers cannot take any step at all
- C) Identical to the standard model's, with robustness coming only from a larger ϵ used at test time
- D) Smoother: small gradients near the correct label and large distances to other labels

ANS:

16. An attacker owns the target model's training set but not its weights. Which strategy does the lecture describe?

- A) Query the target with random inputs until its API returns the gradients of its own loss function
- B) Recover the target's weights by inverting its softmax outputs on the known training data set
- C) Train a proxy white-box model, attack it, and rely on the attacks transferring to the target
- D) Perturb the training set itself, so the deployed target loses accuracy without any transfer

ANS:

17. An attacker has neither the training set nor the weights of a cloud model. What does the lecture suggest, and what is the cost?

- A) Collect input–output pairs by querying the service to train a proxy; this may be expensive
- B) Guess the architecture from the vendor name and train it; the cost is a lower final accuracy
- C) Use the gradients returned by the service API; the cost is zero because gradients are public
- D) Give up, because a usable proxy cannot be built without the original training dataset itself

ANS:

18. How does a Phantom attack on an ADAS differ from the ϵ -ball perturbations earlier in the lecture?

- A) It shifts LiDAR returns by a bounded ϵ so that the range estimates move inside an ℓ^∞ norm ball
- B) It rewrites the ADAS model weights through a firmware update delivered by a roadside radio unit
- C) It adds ℓ_2 -bounded noise to every pixel of every camera frame so humans cannot see it at all
- D) A depthless picture shown for a few frames, with no ℓ_p bound, placed where viewers do not look

ANS:

19. In the objective $\min_{\theta} E[\max_{\delta} \text{Loss}(x+\delta, y; \theta)]$, which gradient updates θ , and what does Danskin's theorem justify?

- A) Inner: maximize over δ to craft the worst-case input; outer: minimize over θ to train on it
- B) Inner: minimize over θ to fit the labels; outer: maximize over δ to craft the attack input here
- C) Inner: the softmax over the logits; outer: the cross-entropy summed over the classes for input x
- D) Both parts are minimizations, one over θ and one over δ , and both are solved at the same time

ANS:

20. In the adversarial-robust objective $\min_{\theta} E[\max_{\delta} \text{Loss}(x+\delta, y; \theta)]$, which part is the inner and which the outer problem?

- A) The gradient of the loss at the clean input x , because the inner max is constant with respect to θ here
- B) The average gradient over all δ in the ball, weighted by the loss that each of those δ produces in turn
- C) The gradient with respect to δ at δ^* , which is zero at the maximizer, so the parameters θ stay unchanged
- D) The gradient of the loss at $x + \delta^*$, with δ^* the exact inner maximizer; the max need not be differentiated

ANS: